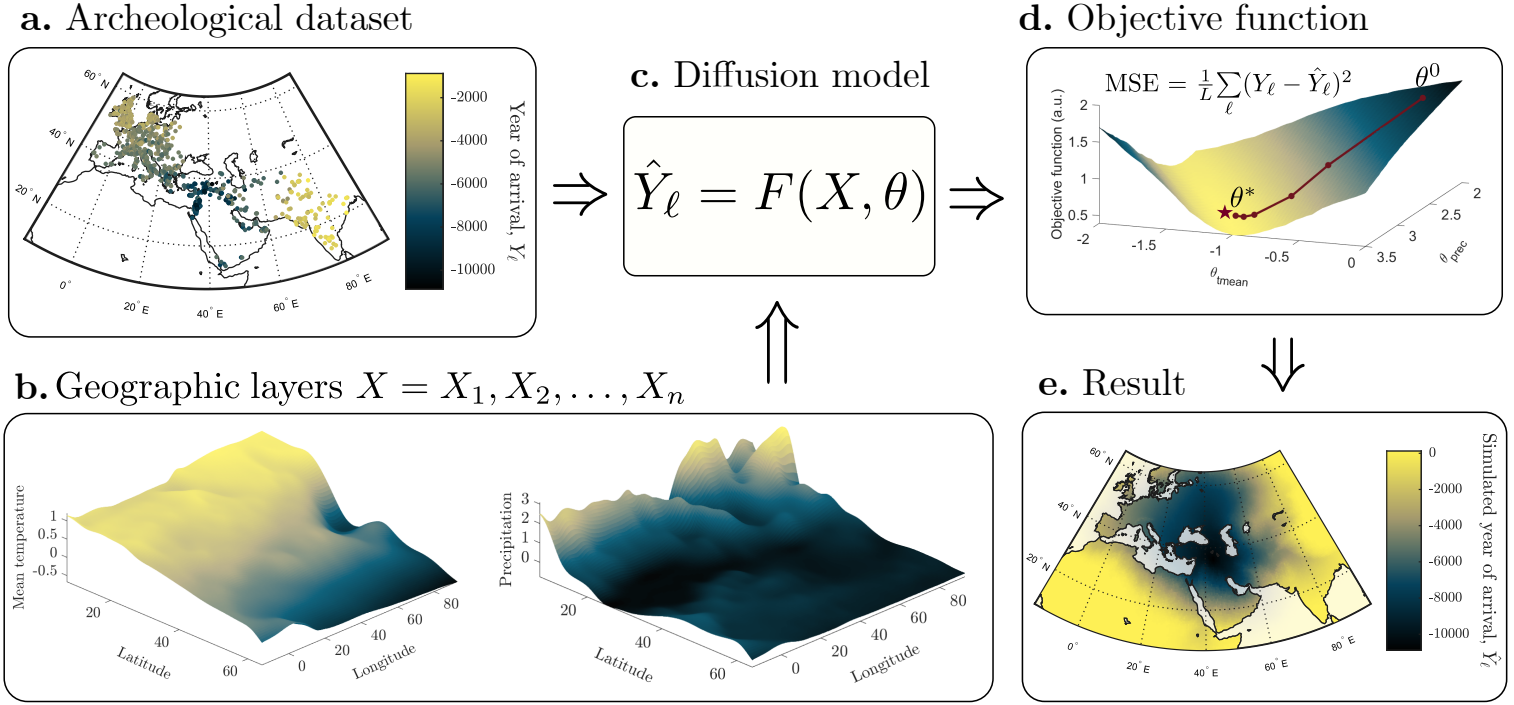
a. Archeological dataset
Year of arrival, Y_ℓ
-2000
-4000
-6000
-8000
-10000
60° N
40° N
20° N
0°
20° E
40° E
60° E
80° E
c. Diffusion model
$\hat{Y}_\ell = F(X, \theta)$
d. Objective function
$\mathrm{MSE} = \frac{1}{L}\sum_\ell (Y_\ell - \hat{Y}_\ell)^2$
θ^0
θ^*
Objective function (a.u.)
θ_{tmean}
θ_{prec}
b. Geographic layers $X = X_1, X_2, \ldots, X_n$
Mean temperature
Precipitation
Latitude
Longitude
e. Result
Simulated year of arrival, Y_ℓ